



Response to the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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Thank you for the opportunity to provide feedback on the Development of an Artificial Intelligence (AI) Action Plan. Given the exciting progress in Artificial Intelligence and the opportunities for profound impacts on our nation, an AI Action Plan is timely and important.

AI is a topic that has arisen from fundamental, curiosity-driven academic research. Many of the recent industry breakthroughs that have brought AI into the mainstream have been driven by a world class workforce of researchers and engineers trained in federally-supported academic labs and universities. The continued health of the field and its future impact require that these investments remain a national priority.

In this reply, we focus on three important actions:

- Invest in building and maintaining the world's most competitive AI workforce
- Prioritize research in high-risk, high-reward innovation
- Strengthen university-industry partnerships to incentivize translational AI research

There are many ways the federal government can act to help continue US dominance in AI, but these three broad topics must be top priority due to their outsize impact and extremely positive return on investment.

Build and maintain the world's most competitive AI workforce

A healthy, well-funded higher education system is critical for maintaining America's competitiveness in AI. Universities, ranging from community colleges to selective institutions, train the majority of AI researchers and engineers through both coursework and hands-on projects. Universities help students develop important numerical, computational, and analytical skills valued by employers; expose students to different cutting-edge application spaces in the physical, biological, and social sciences; and organize and develop crucial K-12 programming, equipping the next generation with the essential tools to contribute to an AI-driven world. History shows that teaching the future top-performers is most effectively accomplished by teachers who themselves are contributing to the cutting edge of the field, necessitating that they have funding support for their work.

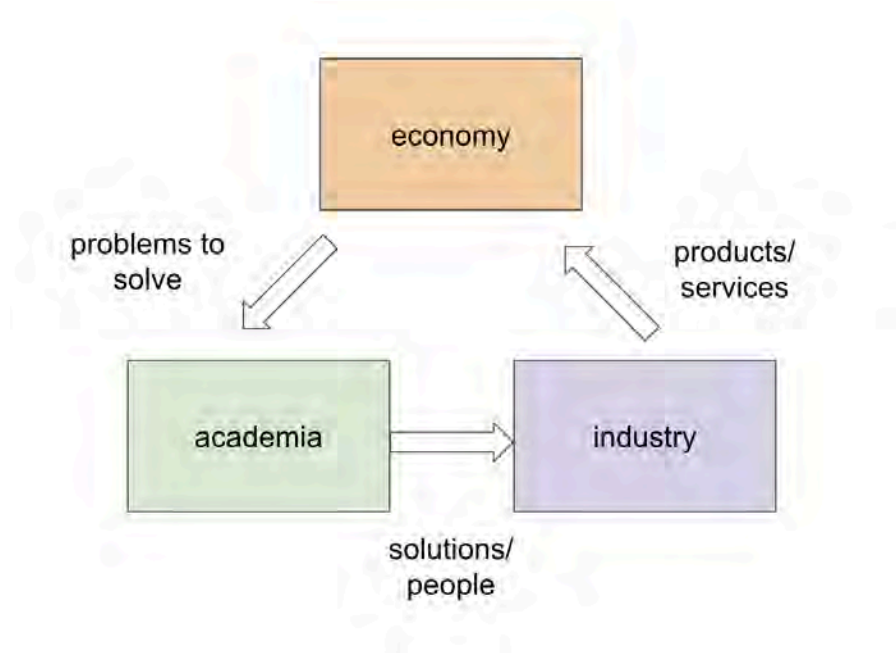


Fig. 1: *Educational Institutions play a critical role in developing an AI workforce; investing in AI education is an investment in the US economy.*

University programs prioritize American students for training. For example, in 2025 at Brown University, approximately 12% of the doctorate-level applicant pool was US Citizens or Permanent Residents yet they represented 65% of all accepted applicants. Even despite our heavy bias toward US Citizens and Permanent Residents, the majority of our graduate students are international. America is known internationally for its esteemed universities and the research that they produce; the world's best and brightest students want to come here and contribute to our R&D efforts.

Specific recommendations:

- **Increased PhD Fellowships for US Citizens:** Establish a National AI Research Graduate Program. The current strong bias towards US Citizens and Permanent Residents is a consequence of the wide availability of fellowship funding for US graduate students. An expansion of these fellowships specifically around AI—NSF GRFP, NSF Grad4CS, and NDSEG, among others, is strongly recommended. A new National AI research graduate student fellowship program would be an effective tool for building the pipeline for American students. Such programs fully fund students to pursue the high-risk, high-reward research that advances the field.
- **Promote AI Literacy in K-12 education:** Develop AI Talent Well Before College. Students must—and can—develop mathematical, analytical, and computational skills as early as in elementary school. It is imperative that national initiatives be undertaken to develop K-12 curricula and programming that foster general analytical, computational, and AI literacy. Programs such as NSF’s “AI4K12” can be supported for implementation nationwide.
- **Teaching the teachers:** Provide Funding for Training Workshops, Programing and Mentorships of K-12 Teachers. Perhaps the biggest challenge in supporting broad AI education is that there are so few teachers with experience presenting this material. The first phase needs to be supporting the education of individuals who can become a workforce of teachers to act as a force multiplier in which they take on the task of training the next level of engineers, subject matter experts, and even more teachers.
- **Motivate the next generation:** Develop Grand Challenges for Students. Science competitions such as Regeneron, the International Math Olympiad, Putnam, the DARPA Grand Challenges on Autonomous Vehicle development, RoboCup soccer challenge, and many others have motivated a generation of scientists to pursue research. Developing grand AI challenges to which students can contribute can at once motivate students and build important skills.
- **Cash-in on American investment in global talent:** Retain Talent. Once trained on how to build advanced AI systems, the world’s best students are often unable to remain in the United States due to their inability to get right-to-remain visas. In addition, international students often enter the country on restrictive visas that do not allow re-entry to the country. This limits academia’s ability to present and share our research on the world stage, at academic conferences and other global research meetings. Talent retention is critical for progress and a visa pathway for talent to remain in the US is recommended.

Prioritize research in high-risk, high-reward innovation

While industry invests in research that will pay off within 5 years, academia keeps its focus on research that will pay off within 10-15-20 years, or even longer. This creates an innovation pipeline for America. Academic research, which supports early stage, high-risk research, provides the foundation for downstream industry breakthroughs—reduction in academic education and research will hamper both near term and long term AI innovation. Note that early investments in statistical machine

translation from the US government provided both the technical foundations and human resources on top of which Google Translate was founded in 2006. These investments later produced the core technologies such as Transformer neural networks that underlie the current AI revolution. In addition, profitability incentives encourage industry to focus more on scaling to best cater to customers' needs, while academia has the freedom to explore the foundations untethered to immediate revenue outcomes, enabling innovation that can yield truly transformative pre-competitive insights that would fall outside of any given industry or product sector today.

Specific recommendations:

- **High-Risk, High-Reward Research:** Create Blue Sky Funding Opportunities for Moonshot Projects. High-risk, high-reward research has time-and-again led to transformative advances; however, a limited amount of funding is currently dedicated to such research. We advocate establishing sufficient long-term funding for blue sky, trailblazing research. We would expect many such projects to fail, but that the ones that succeed would have outsized impact. (Foundational research is commonly estimated to have a 5-to-1 return on investment.)
- **Opening the Blackbox:** Invest in AI Research to “Open Up the Black Box” of AI, Investigation, including but not limited to work on AI theory and interpretability, will improve products and applications in the long run—say, by making them easier to control, more efficient to build, and/or more adaptable to deploy—but not in the timeline on which industry investments are typically measured.
- **AI Incubators:** Bring Together Universities, Industry, and Community Partners to Solve Local Grand Challenges. AI incubators that large companies can then scale to the national level are a royal road to transformative new developments. For example, local regions often have specific medical and population needs that may not be appreciated on the national scale; it is often additionally challenging to share that data due to a variety of protections. Local AI incubators would thus be particularly well-suited for addressing local grand challenges, in areas such as cancer, transportation, or education.
- **AI for Science:** Invest in the Intelligent Acquisition of Experimental Data in the Physical, Biological, and Social Sciences. Data drives a variety of important AI applications and individual research labs play a critical role in collecting small, but costly data that can be aggregated into larger, more general data sets that are then amenable to industrial or other uses. One striking example is the generation of the Protein Databank (PDB), which was critical to AlphaFold's success; the PDB's data was generated through thousands of experiments by independent university (and other) research groups over decades. The government plays a pivotal role in organizing the acquisition and sharing of such data.
- **Resources and Infrastructure:** Provide Access to Large Scale Computing Resources for Researchers. Modern AI research is immensely compute and data intensive, and as a result many research questions can only be studied in well-resourced labs. This results in a massively inefficient use of intellectual and research effort outside of the richest and most established labs. Establishing national research infrastructure, such as the National AI Research Resource

(NAIRR), would remove a significant barrier to innovation, engaging the nation's sharpest minds, wherever they are.

Strengthen university-industry partnerships to incentivize translational AI research

Universities play a crucial role in research and innovation that is distinct from, though ultimately enables, the innovation of industry. For example, the current AI revolution has depended on several key innovations described in high impact papers that originated in university ecosystems. Among these are the idea of the *Transformer architecture* which allows for training very large AI models, the idea of *Instruction Tuning* which allows for AI models to be responsive to the needs of non-technical users, and the idea of *AlphaFold* which uses AI to make fundamental advances in biology and medicine (and resulted in a Nobel Prize). The research papers behind these advances were first authored by Asish Vaswani (PhD USC), Long Ouyang (PhD Stanford), and John Jumper (PhD University of Chicago), respectively, all of whom were trained in federally funded university research labs. Increasing investment in the interface between universities and industry will strengthen the United State's unique competitive advantage that comes from having both world-class higher education and an entrepreneurial economy that is the envy of all.

Specific recommendations:

- **Open Source AI:** Protect Open Source and Open Science. Both American universities and American companies benefit from a strong open source and, more broadly, open science culture. There have been discussions in Congress about restrictions due, mostly, to competition with China—for example, some restrictions have been discussed in the ENFORCE bill proposals from the last Congress. Restricting open source and open science would have a chilling effect on both academic research and American innovation; we ask the administration to protect open source and open science.
- **Standards:** Develop Reproducibility Standards for Published AI Research. Computational scientists have been impelled to openly share their data in transparent and reproducible ways (FAIR principles) upon publication. We strongly encourage the adoption of similar transparency and reproducibility standards for AI models.
- **Professional Development:** Create Career Development Pathways for Industry Professionals. Upskilling and reskilling are key for the current workforce and institutions of higher education can help. Further, industry has a key role to play in identifying skills and topics that can be taught in school to prepare students for the landscape of today.
- **Continuous Executive Education:** Establish an Executive Fellowship. We advocate for a bidirectional opportunity for academics to work in industry to address immediate challenges, and for industry experts to receive a temporary academic position to explore early stage research questions.
- **Industry Investment:** Incentivize Industry Investments in Science. Industry currently supports a small number of students through highly competitive

fellowships. These opportunities could be expanded to support many more students. This encourages the development of industry research agendas during the high-risk, high reward phase of development that occurs in academic environments. Incentivizing industry to participate in such industry investment should supplement and not reduce federal investment in science due to the divergence of public interest innovation and industry innovation.

These recommendations constitute a vital set of actions the federal government can take to move the Nation's advantages in artificial intelligence to the next level, resulting in massive benefits for the economy and Americans broadly.